

Association-Based Fairness Measures under Class and Group Skew

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Fairness audits in binary classification assess the differences between protected and unprotected groups, using measures such as statistical parity, equal opportunity, predictive equality, and predictive parity. However, prior work on algorithmic fairness has shown that several of these measures exhibit substantial behaviour changes as class imbalance and protected-group ratios vary. This paper presents ongoing work on whether association-based measures can provide more stable fairness evaluation under three forms of skew: class imbalance, protected-group ratio, and stereotypical ratio. In this context, we adapt classical association coefficients to fairness tables and propose three new metrics: Fairness ϕ , Marginal Y Association, and Conditional Y Association. The first two measure independence between the prediction and the protected attribute, whereas the third aggregates within-label associations and is intended to reflect separation-style disparities. We study these metrics on synthetic confusion matrices and empirically on the Adult dataset. Preliminary results indicate that the association-based metrics, in particular Fairness ϕ and Conditional Y Association, are less sensitive to extreme class imbalance and protected-group skew. The trade-off is that the definitions of these measures are harder to interpret on a conceptual level and, for the conditional variant, require smoothing. Nevertheless, these results indicate that association-based fairness metrics may be useful audit-oriented tools when comparing models across skewed datasets.

Keywords: algorithmic fairness, fairness metrics, class imbalance, protected-group imbalance, stereotypical bias

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1 Introduction

Fairness evaluation plays a central role in machine learning systems used in socially consequential settings, yet there is no single metric that is correct across all contexts [2, 13]. In binary classification, group fairness is usually defined through parity of selection rates, error rates, or predictive values between groups. These measures are easy to report but are not interchangeable, and several of them are mutually incompatible [7, 10]. The problem becomes more pronounced when the data are skewed. Class imbalance, protected-group imbalance, and label-conditional effects can all change how a fairness metric behaves, even when the underlying predictive model is fixed [5, 8, 11].

Prior studies [5] have shown that standard group fairness measures can differ in stability under changing imbalance ratios and group proportions. In particular, accuracy equality and statistical parity were relatively stable under varying conditions, whereas equal opportunity, predictive equality, and predictive-parity measures were more

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sensitive and often undefined in scenarios of extreme skew. This raises a measurement problem: if the metric itself is unstable, fairness comparisons across deployment settings may become incomparable [4].

Here, we adopt the framework presented in [5] for investigating stability under varying imbalance and group ratios with the aim of identifying new measures that yield more stable evaluations under skewed conditions. Instead of modifying existing fairness metrics, we propose using association-based approaches known from statistical tests for 2×2 contingency tables. The goal is to test whether adaptations of such association measures are more stable than popular independence and separation fairness metrics under changing class, protected-group, and label-conditional skew. As a result, we put forward three new association-inspired fairness metrics: Fairness ϕ , Marginal Y Association, and Conditional Y Association. We then compare the behaviour of the newly proposed measures with standard metrics under the same skew-oriented experimental protocol.

2 From Association Measures to Fairness Metrics

Most common group fairness measures in binary classification belong to the independence or separation families and are expressed as differences or ratios [6, 16]. Here, we adapt classical statistical association measures for binary contingency tables to the fairness setting. The proposal is inspired by Pearson’s ϕ coefficient and Yule’s Y association measure for 2×2 tables [1, 14, 15]. Using association measures allows us to measure how strongly the whole prediction table is associated with group membership, rather than tracking one selected class or group.

Let $A \in \{i, j\}$ be the protected attribute and $\hat{Y} \in \{0, 1\}$ the prediction. We will define association measures based on the following prediction count table:

	$\hat{Y} = 1$	$\hat{Y} = 0$
$A = j$	n_{11}	n_{01}
$A = i$	n_{10}	n_{00}

where, for example, $n_{11} = N(A = j, \hat{Y} = 1)$ is the number of predictions of (positive) class 1 for examples belonging to (protected) group j . Based on this table, we propose two signed measures derived from Pearson’s ϕ and Yule’s Y coefficients:

$$\text{Fairness } \phi = \frac{n_{11}n_{00} - n_{10}n_{01}}{\sqrt{n_{1\cdot}n_{0\cdot}n_{\cdot 1}n_{\cdot 0}}}, \quad \text{Marginal } Y = \frac{\sqrt{n_{11}n_{00}} - \sqrt{n_{10}n_{01}}}{\sqrt{n_{11}n_{00}} + \sqrt{n_{10}n_{01}}}.$$

Both are zero when the prediction is independent of the group ($\hat{Y} \perp A$). Therefore, these measures belong to the family of *independence* fairness metrics.

The third proposed measure conditions on the true class label Y and belongs to the family of *separation* fairness metrics, i.e., it is zero when $\hat{Y} \perp A \mid Y$. Let Y_0 and Y_1 denote Yule’s Y (computed similarly to Marginal Y), but computed only for examples with true class labels $Y = 0$ and $Y = 1$, respectively. We aggregate these two within-class associations (Y_0, Y_1) using the root mean square to obtain a single measure:

$$\text{Conditional } Y = \sqrt{\frac{Y_0^2 + Y_1^2}{2}}.$$

We use the root mean square because the Y coefficients for the two class labels can have opposite signs; thus, a simple average could cancel them out, making a conditionally unfair model appear fair. The root mean square avoids this cancellation, treats both classes symmetrically, and preserves a magnitude-style interpretation. In the

implementation, this conditional measure uses $+0.5$ Haldane–Anscombe smoothing to avoid degenerate odds-ratio calculations [12].

We do not define a conditional analogue of Fairness ϕ , because unlike the Y coefficient, ϕ is not directly determined by the within-class odds ratio. As a result, aggregating ϕ across the $Y = 0$ and $Y = 1$ strata would not preserve the same conditional-association interpretation.

3 Experiments and Preliminary Findings

We experimentally evaluated the proposed measures in two stages as in [5] (Figure 1). First, we studied dataset-independent behaviour by analysing distributions of metric values for every possible confusion matrix at $n = 30$. We then performed a controlled empirical protocol using the UCI Adult dataset [3], with sex as the protected attribute. The proposed measures were compared against seven standard group fairness measures: Accuracy Equality Difference, Statistical Parity Difference, Equal Opportunity Difference, Equalized Odds Difference, Predictive Equality Difference, Positive Predictive Parity Difference, and Negative Predictive Parity Difference. The code and data for reproducing the experiments are available at <https://github.com/dromaniv/fairness-measures-under-skew>.

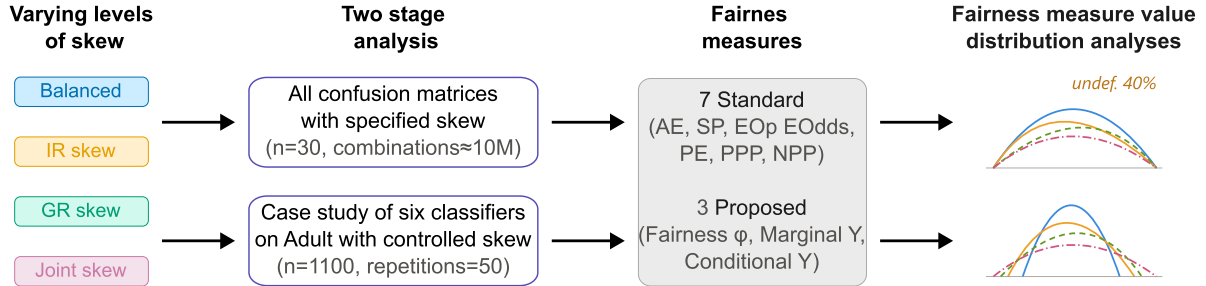


Fig. 1. Overview of the experimental evaluation of fairness measures under varying skew conditions defined by IR, GR, and SR. The distributions of fairness measures were analysed in two stages: 1) a dataset-independent analysis that enumerates all possible confusion matrices ($n = 30$) with expected skew and 2) a six-classifier case study on UCI Adult with controlled skews ($n = 1,100$; 50 repetitions of holdout tests).

In the experiments, we define the two main forms of skew by the class imbalance ratio (IR) and the protected-group ratio (GR):

$$IR = \frac{P}{n}, \quad GR = \frac{n_j}{n},$$

where P is the number of positive examples ($Y = 1$), n_j is the size of the protected group j , and n is the dataset size. We then introduce a third notion of skew to address stereotypical bias [11]. Let SR_p be the fraction of all positives that are in the protected group j , SR_n the fraction of negatives in group j , and SR_c their geometric mean:

$$SR_p = \frac{P_j}{P}, \quad SR_n = \frac{N_j}{N}, \quad SR_c = \sqrt{SR_p SR_n}.$$

These ratios are motivated by the observation that overall demographic balance can hide label-conditional skew [11]. Figures 2 and 3 present the experimental results of both evaluation stages on a sample of four conditions: a balanced reference ($IR = 0.5, GR = 0.5$), class-imbalance skew ($IR = 0.1, GR = 0.5$), group-ratio skew ($IR =$

0.5, $GR = 0.1$), and joint skew ($IR = 0.1, GR = 0.1$). For Figure 2, the curves are built by computing each metric on all possible confusion matrices in the selected condition and binning the finite values into equal-width intervals. If present, undefined values are excluded from the curves and reported separately in coloured italic annotations. Figure 3 follows the same visual format, but its distributions come from averaged outputs of six classifiers (Random Forest, MLP, Naive Bayes, k-NN, Decision Tree, Logistic Regression) over repeated Adult dataset holdout splits.

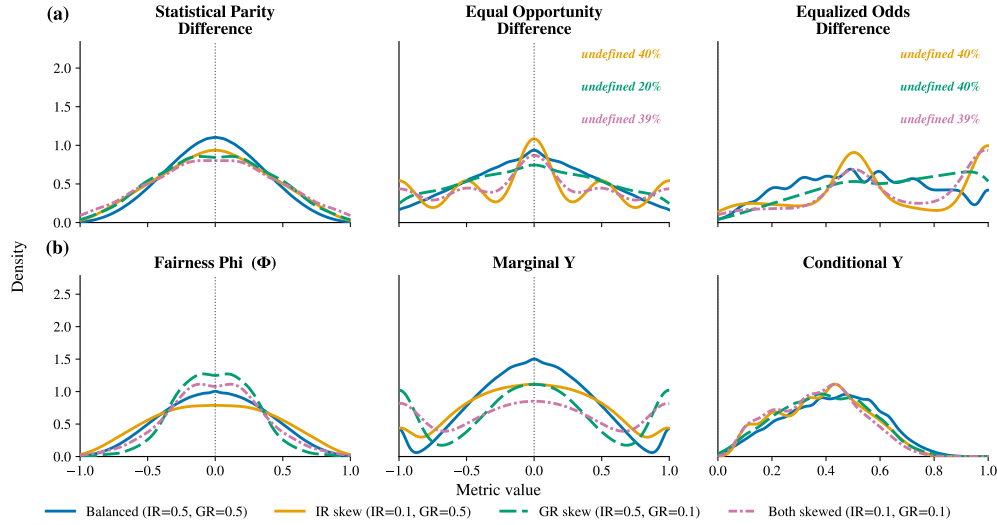


Fig. 2. Distributions of representative fairness metrics over fully enumerated confusion matrices ($n = 30$) under four conditions of class-imbalance (IR) and group-ratio (GR) skew. Row (a) shows standard metrics; row (b) shows the proposed association-based metrics. Italic annotations report the fraction of confusion matrices for which a metric is undefined. It can be noticed that Equal Opportunity and Equalized Odds Difference accumulate undefined values up to 40% and Statistical Parity Difference disperses under group-ratio skew, whereas the proposed metrics remain defined and concentrated across all four conditions.

Three patterns surface in the synthetic enumeration and recur in the case study. (i) Equal Opportunity Difference and Equalized Odds Difference accumulate up to $\sim 40\%$ undefined values under class imbalance, and the surviving distributions are irregular and multi-modal; on Adult, both metrics lose peak height as soon as either marginal distribution becomes skewed. (ii) Statistical Parity Difference is roughly insensitive to class imbalance, but disperses under group-ratio and joint skew; the same is visible on Adult. (iii) Fairness ϕ remains concentrated near zero across all four conditions, whereas Statistical Parity Difference visibly broadens once the group ratio becomes skewed. Marginal Y Association varies across conditions, but less sharply than, e.g., Equalized Odds. Conditional Y Association remains defined throughout and shifts only modestly. Preliminary results for varying SR suggest that SR has a similar effect to GR , and that Conditional Y Association remains largely insensitive to SR skew.

4 Discussion

The current evidence suggests that selected association-based coefficients, when adapted to fairness tables, can serve as more skew-resistant complements to standard fairness metrics. However, these measures are less directly

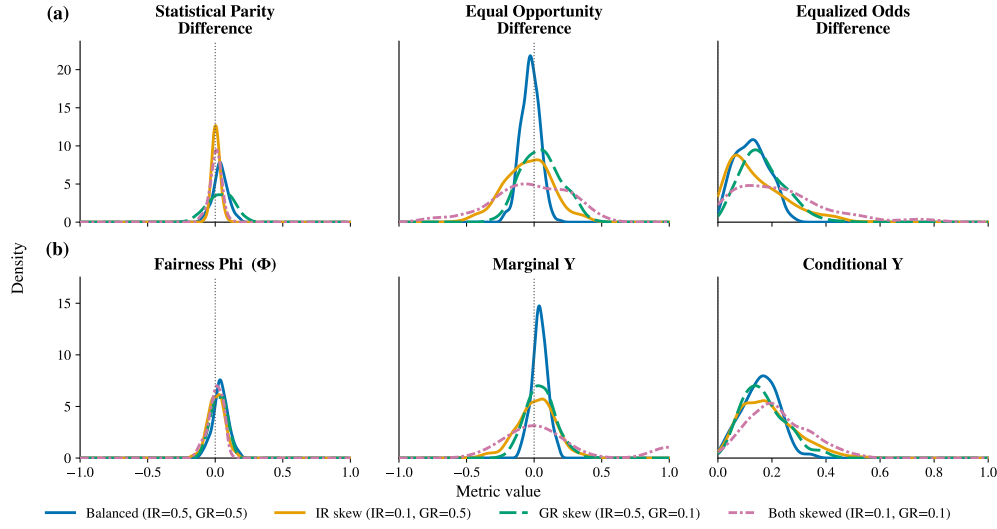


Fig. 3. Case study on the UCI Adult dataset. Distributions are obtained from six classifiers evaluated over 50 stratified holdout splits ($n = 1,100$ per split, 33% held out; protected attribute: sex; positive class: income $> 50K$). As was the case for confusion-matrix analyses presented in Figure 2, in model evaluation on a real dataset, the standard metrics disperse in the presence of skews, whereas Fairness ϕ 's four densities are nearly indistinguishable across conditions.

tied to standard parity constraints used in the fairness literature, and the conditional variants require smoothing and therefore require more careful interpretation.

The next step is to extend the synthetic and Adult experiments and check whether the patterns hold across more experimental settings, such as fairness post-processing techniques for classifier predictions [9]. Another avenue for future research is to investigate whether an analogous association-based metric can be constructed for the sufficiency family, to complement the current independence- and separation-oriented proposals. If the patterns persist, fairness audits could benefit from metrics that remain defined, bounded, and stable under skewed data. This matters for both benchmarking and real-world audits, where datasets rarely share the same prevalence or group composition.

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